PROBLEM 5 · PROBLEM STATEMENT

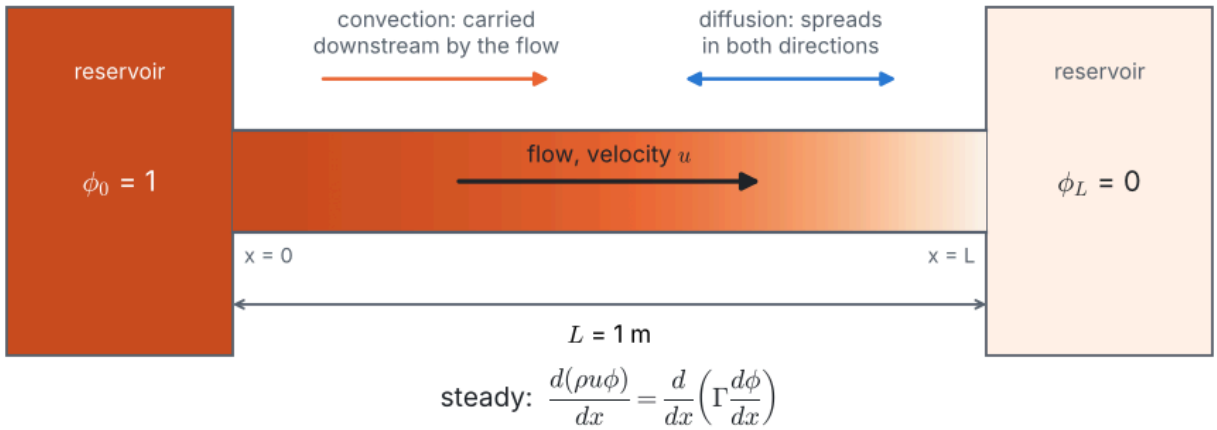
Steady convection and diffusion in one dimension

Central versus upwind differences for the convection term, solved with the Finite Difference Method

The problem

A property ϕ (for example a temperature or the concentration of a dye) is **transported by convection and diffusion** through a one-dimensional domain $0 \le x \le L$ between two reservoirs. The fluid flows from left to right with a **constant velocity** u. The left reservoir holds $\phi = 1$ and the right reservoir holds $\phi = 0$. Find the **steady-state distribution of** ϕ with the **Finite Difference Method**, discretizing the convection term with **central differences** and with **upwind differences**, for a slow flow ($u = 0.1$ m/s) and a fast flow ($u = 2.5$ m/s).

(Adapted from Versteeg & Malalasekera, An Introduction to Computational Fluid Dynamics, chapter 5.)



A fluid flows with velocity u from a reservoir at $\phi_0 = 1$ to a reservoir at $\phi_L = 0$. Convection carries ϕ downstream; diffusion spreads it in both directions.

Governing equation

In steady state, with no source of ϕ:

$$\frac{d(\rho u\, \phi)}{dx} = \frac{d}{dx}\left(\Gamma \frac{d\phi}{dx}\right), \qquad \phi(0) = \phi_0 = 1, \qquad \phi(L) = \phi_L = 0.$$

The flow also satisfies continuity, $d(\rho u)/dx = 0$, so ρu is constant.

Data

Quantity	Symbol	Value
length of the domain	L	1.0 m
density	ρ	1.0 kg/m³
diffusion coefficient	Γ	0.1 kg/(m·s)
value at $x = 0$ (left reservoir)	ϕ_0	1
value at $x = L$ (right reservoir)	ϕ_L	0
velocity, case 1 (slow flow)	u	0.1 m/s
velocity, case 2 (fast flow)	u	2.5 m/s
grid for cases 1 and 2	N, Δx	5 intervals, $\Delta x = 0.2$ m (nodes 0 to 5, unknowns 1 to 4)
grids for case 3 (refinement)	N	5, 10, 20, 40, 80, 160, 320 intervals, $u = 2.5$ m/s

The exact solution of this problem is

$$\frac{\phi(x) - \phi_0}{\phi_L - \phi_0} = \frac{\exp(\rho u x/\Gamma) - 1}{\exp(\rho u L/\Gamma) - 1}.$$

Data for simulation software (for example COMSOL Multiphysics, *Transport of Diluted Species*, 1D):

- geometry: an interval from $x = 0$ to $x = 1$ m;
- diffusion coefficient $D = \Gamma/\rho = 0.1$ m²/s; convection by a given velocity field u (no flow equations need to be solved);
- velocity $u = 0.1$ m/s (case 1) and $u = 2.5$ m/s (cases 2 and 3);
- boundary conditions: concentration $c(0) = 1$ mol/m³ and $c(L) = 0$; no reactions or sources; stationary study;
- mesh: uniform elements of size 0.2 m (5 elements) for cases 1 and 2; 0.1, 0.05, 0.025, 0.0125, 0.00625 and 0.003125 m (10 to 320 elements) for case 3.

With c in mol/m³ the numbers are the same as ϕ. Commercial codes add numerical **stabilisation** to the convection term by default; to compare with the plain central-difference scheme, switch it off.

Assumptions

- Steady state: ϕ no longer changes with time.
- One-dimensional transport along x.
- Constant density ρ, velocity u and diffusion coefficient Γ.
- No source or sink of ϕ inside the domain.

Tasks

1. **Physics.** Explain the two terms of the equation: what does convection do, and what does diffusion do? Compute the **Péclet number** $Pe_L = \rho u L/\Gamma$ for both velocities. Sketch how the exact solution changes as u grows.

2. **Grid.** Divide the domain into $N = 5$ equal intervals. Compute the **cell Péclet number** $Pe_\Delta = \rho u \, \Delta x / \Gamma$ for both velocities.
3. **Discretization.** Use a central difference for the diffusion term, and for the convection term
   a. a **central difference** (CDS): $d\phi/dx|_i \approx (\phi_{i+1} - \phi_{i-1})/(2\Delta x)$;
   b. an **upwind difference** (UDS): $d\phi/dx|_i \approx (\phi_i - \phi_{i-1})/\Delta x$ for $u > 0$.

   Write each interior equation in the form $a_P \, \phi_i = a_W \, \phi_{i-1} + a_E \, \phi_{i+1}$ and give a_W, a_E and a_P in terms of $F = \rho u$ and $D = \Gamma / \Delta x$.
4. **Solution.** Solve the tridiagonal systems with the **Thomas algorithm**.
5. **Case 1** ($u = 0.1$ m/s) and **case 2** ($u = 2.5$ m/s). Compare CDS, UDS and the exact solution at the nodes. Which results are physically possible? Explain what you see using the signs of the coefficients.
6. **False diffusion.** Show that the upwind scheme equals the central scheme with an extra diffusion coefficient $\rho u \, \Delta x / 2$. How large is it compared with Γ?
7. **Case 3, verification.** For $u = 2.5$ m/s, refine the grid ($N = 5$ to 320). Plot the largest error against Δx on log–log axes for both schemes. When does CDS start to work? What are the orders of accuracy?
8. *Optional.* Solve the case-2 equations with **Gauss–Seidel** iteration. What happens with CDS?

What you will learn

- What the **convection term** does, and why it is what makes CFD *fluid* dynamics.
- What the **Péclet number** measures, and why the **cell** Péclet number decides how a scheme behaves.
- Why central differences produce **non-physical wiggles** when $Pe_\Delta > 2$ (negative coefficients, loss of diagonal dominance), and why upwinding is always **bounded**.
- What **false diffusion** is, and why commercial codes use upwind-biased or stabilised schemes.
- How a grid-refinement study reveals the **order of accuracy** of a scheme.

> **Files for this problem:** `02_solution.pdf` explains the solution step by step. `03_fdm_solver.ipynb` contains the complete Python code (plain Python, no numerical libraries), with explanations and plots, to run cell by cell. `04_whiteboard_guide.pdf` solves cases 1 and 2 by hand for the board.